

Question(s) for rec08 - Separate Compilation

Given the following three files:

```
// a.h
#include "b.h" // Line A
```

```
// b.h
#include "a.h" // Line B
```

```
// c.cpp
#include "a.h" // Line C
#include <iostream>

int main() {
    std::cout << "Hm...\n";
}
```

What will be the result of building and running?

- | | |
|--|---|
| ☐ Compilation error at line A | ☐ Undefined compilation error |
| ☐ Compilation error at line B | ☐ Displays: Hm... |
| ☐ Compilation error at line C | ☐ Displays a blank line |
| ☐ Fails to compile because c.cpp
does not include b.h | ☐ None of the above |

Given the following three files:

```
// a.h
#ifndef A_H
#define A_H
#include "b.h"
struct A {
    B* bp; // Line A
};
#endif
```

```
// b.h
#ifndef B_H
#define B_H
#include "a.h"
struct B {
    A* ap; // Line B
};
#endif
```

```
// c.cpp
#include "a.h" // Line C
#include <iostream>

int main() {
    std::cout << "Hm...\n";
}
```

What will be the result of building and running?

- ☐ Compilation error at line A
- ☐ Compilation error at line B
- ☐ Compilation error at line C
- ☐ Fails to compile because c.cpp does not include b.h
- ☐ Displays: Hm...
- ☐ Displays a blank line
- ☐ Undefined output
- ☐ None of the above